PES UNIVERSITY

Computer Organization and Design

(UE23EC352A)

Application using GSM-Example:Fire detection and Alerting System

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Introduction

A fire detection and alerting system is a critical safety application that combines fire sensors with GSM (Global System for Mobile Communications) technology to detect fire incidents and automatically send alert messages to registered mobile numbers. The system uses flame or smoke sensors to detect fire, processes the signal through a microcontroller, and triggers a GSM module to send SMS alerts to emergency contacts, fire departments, or building occupants. This real-time notification system enables rapid response to fire emergencies, potentially saving lives and minimizing property damage.

Applications:

Residential Buildings: Automated fire alerts to homeowners and fire departments
Industrial Facilities: Early warning system for warehouses and manufacturing plants
Forests: Wildfire detection in remote areas where traditional monitoring is difficult
Commercial Buildings: Shopping malls, offices, and hotels for occupant safety
Agricultural Storage: Protection of grain silos and storage facilities
Parking Garages: Detection in enclosed vehicle storage areas
Laboratories: Protection of research facilities with hazardous materials
Server Rooms: Protection of critical IT infrastructure

Objectives

1. Implement RISC-V assembly fire detection with temp, smoke, flame sensors
2. Develop smart multi-condition logic (2+ triggers) to reduce false alarms
3. Integrate GPS coordinates for emergency location reporting
4. Display comprehensive alerts with sensor data and emergency contacts
5. Create resource-efficient system optimized for embedded platforms

Tools Used

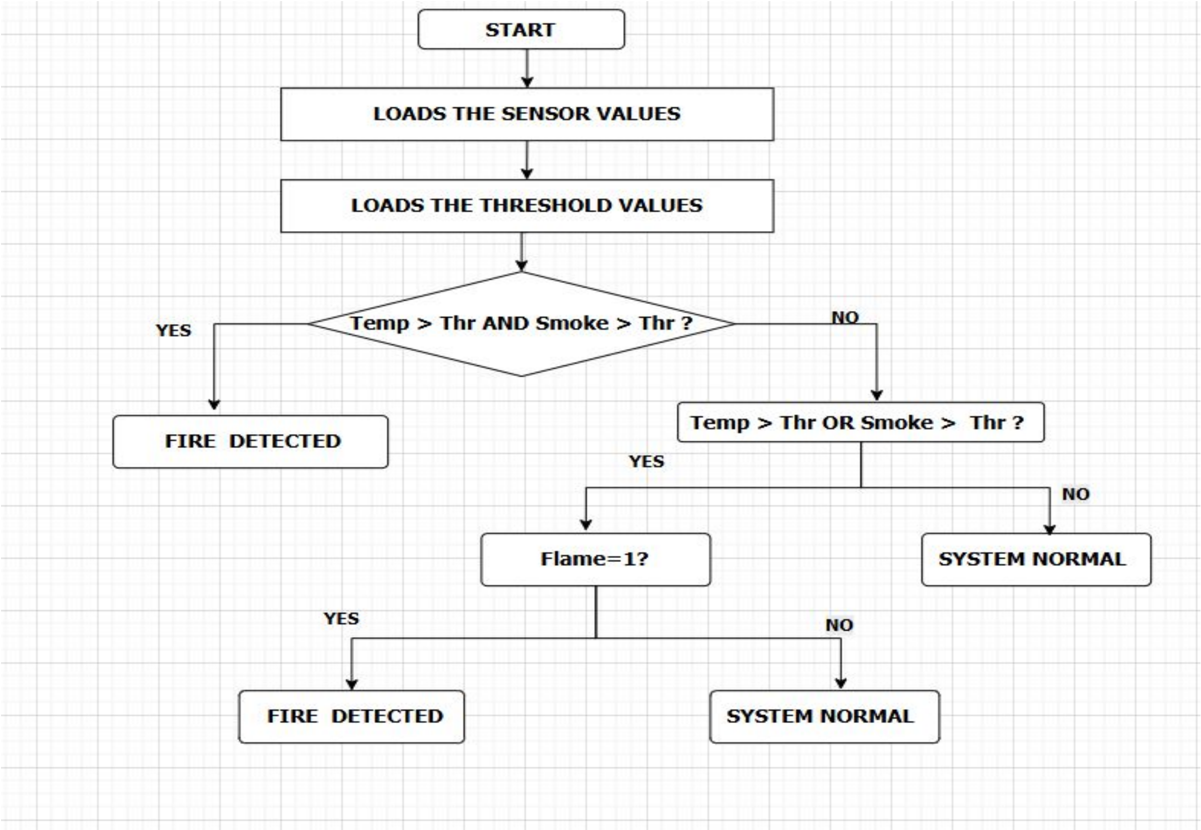
Ripes Simulator – A RISC-V simulator used to write, assemble, and execute the fire detection program.

- Simulates CPU, memory, and registers.
- Sensor values (temp_sensor, smoke_sensor, flame_sensor) are simulated via .data.
- Fire alerts and contact messages are displayed on the console.

Tools Used (Real-World Implementation)

- RISC-V Development Board
- Temperature, Smoke, and Flame Sensors
- GSM Module with SIM Card (for alerts)
- LED / Buzzer (optional)
- Wires, Breadboard, and Power Supply

Flowchart


START
LOADS THE SENSOR VALUES
LOADS THE THRESHOLD VALUES
Temp > Thr AND Smoke > Thr ?
YES
NO
FIRE DETECTED
Temp > Thr OR Smoke > Thr ?
YES
NO
Flame=1?
SYSTEM NORMAL
YES
NO
FIRE DETECTED
SYSTEM NORMAL

Code

```
.data
# -------- Sensors --------
temp_sensor:        .word 55          # Temperature sensor value (integer)
smoke_sensor:       .word 600          # Smoke sensor value (integer)
flame_sensor:       .word 0            # Flame sensor (1 = Yes, 0 = No)

# -------- Thresholds --------
temp_thr:           .word 45           # Temperature threshold
smoke_thr:          .word 500          # Smoke threshold

# -------- GPS Coordinates (example integers in microdegrees) --------
gps_lat:            .word 12971600     # Latitude 12.9716° stored as 12.9716*1e6
gps_long:           .word 77594600     # Longitude 77.5946° stored as 77.5946*1e6

# -------- Messages --------
fire_msg:           .string "FIRE DETECTED!\n"          # Fire alert message
contact_msg:        .string "Call Fire Dept: 101\n"    # Fire dept contact
owner_msg:          .string "Owner Contact: +91-9876543210\n" # Owner contact
normal_msg:         .string "SYSTEM NORMAL\n"          # Normal system message
temp_msg:           .string "Temperature: "            # Label for temp print
smoke_msg:          .string "Smoke: "                  # Label for smoke print
flame_msg:          .string "Flame: "                  # Label for flame print
lat_msg:            .string "Latitude: "               # Label for latitude print
long_msg:           .string "Longitude: "              # Label for longitude print
yes_str:            .string "Yes\n"                    # Flame = 1
no_str:             .string "No\n"                     # Flame = 0
.text
.globl main
main:
    # -------- Load Sensor Values --------
    la t0, temp_sensor      # Load address of temp_sensor into t0
    lw t1, 0(t0)            # Load temp_sensor value into t1

    la t0, smoke_sensor
    lw t2, 0(t0)            # Load smoke_sensor value into t2

    la t0, flame_sensor
    lw t3, 0(t0)            # Load flame_sensor value into t3

    # -------- Load Thresholds --------
    la t0, temp_thr
    lw t4, 0(t0)            # Load temperature threshold into t4

    la t0, smoke_thr
    lw t5, 0(t0)            # Load smoke threshold into t5

# -------- FIRE LOGIC (2+ conditions) --------
    blt t4, t1, TEMP_HIGH  # If temp > threshold, branch to TEMP_HIGH
    j CHECK_FLAME           # Else check flame combination

TEMP_HIGH:
    blt t5, t2, FIRE        # If smoke > threshold → FIRE
    j CHECK_FLAME           # Else check flame

CHECK_FLAME:
    bne t3, x0, FLAME_COMBO # If flame sensor != 0 → check combination
    j NORMAL                # Else system is normal

FLAME_COMBO:
    blt t4, t1, FIRE        # Flame + temp > threshold → FIRE
    blt t5, t2, FIRE        # Flame + smoke > threshold → FIRE
    j NORMAL                # Flame alone → SYSTEM NORMAL

# -------- NORMAL MODE --------
NORMAL:
    la a0, normal_msg       # Load string address
    li a7, 4                # ecall code 4 → print string
    ecall
    j END                   # End program

# -------- FIRE ALERT MODE --------
FIRE:
    la a0, fire_msg
    li a7, 4                # ecall 4 → print string
    ecall

    # Print Temperature
    la a0, temp_msg
    li a7, 4
    ecall
    mv a0, t1               # Move temperature value to a0
    li a7, 1                # ecall 1 → print integer
    ecall
    li a0, 10
    li a7, 11               # ecall 11 → newline
    ecall

    # Print Smoke
    la a0, smoke_msg
    li a7, 4
    ecall
    mv a0, t2
    li a7, 1
    ecall
    li a0, 10
    li a7, 11
    ecall
```

Code

```
    # Print Flame
    la a0, flame_msg
    li a7, 4
    ecall
    bne t3, x0, FLAME_YES     # If flame = 1
    la a0, no_str              # Else print "No"
    li a7, 4
    ecall
    j FLAME_END

FLAME_YES:
    la a0, yes_str             # Print "Yes" if flame = 1
    li a7, 4
    ecall

FLAME_END:

    # -------- Print GPS Coordinates --------
    la t0, gps_lat
    lw a0, 0(t0)              # Load latitude into a0
    la t1, lat_msg
    mv a0, a0
    la a0, lat_msg
    li a7, 4                  # Print label
    ecall
    la t0, gps_lat
    lw a0, 0(t0)              # Print latitude value
    li a7, 1
    ecall
    li a0, 10
    li a7, 11                 # Newline
    ecall

    la t0, gps_long
    lw a0, 0(t0)              # Load longitude into a0
    la a0, long_msg
    li a7, 4                  # Print label
    ecall
    la t0, gps_long
    lw a0, 0(t0)              # Print longitude value
    li a7, 1
    ecall
    li a0, 10
    li a7, 11                 # Newline
    ecall

```

```

    # Print Contact Info
    la a0, contact_msg
    li a7, 4
    ecall
    la a0, owner_msg
    li a7, 4
    ecall

# -------- END PROGRAM --------
END:
    li a7, 10                 # ecall 10 → exit
    ecall

```

Results

Test Case	Temp (°C)	Smoke (PPM)	Flame	Expected	Actual	Status
1	25	150	0	Normal	Normal	✓ Pass
2	55	600	0	Fire	Fire	✓ Pass
3	60	200	1	Fire	Fire	✓ Pass
4	30	550	1	Fire	Fire	✓ Pass
5	30	200	1	Normal	Normal	✓ Pass
6	50	200	0	Normal	Normal	✓ Pass
7	30	550	0	Normal	Normal	✓ Pass

```
# -------- Sensors --------
temp_sensor:        .word 25
smoke_sensor:       .word 150
flame_sensor:       .word 0
```

```
SYSTEM NORMAL

Program exited with code: 0
```

```
# -------- Sensors --------
temp_sensor:        .word 55
smoke_sensor:       .word 600
flame_sensor:       .word 0

FIRE DETECTED!
Temperature: 55
Smoke: 600
Flame: No
Latitude: 12971600
Longitude: 77594600
Call Fire Dept: 101
Owner Contact: +91-9876543210

Program exited with code: 0
```

```
# -------- Sensors --------
temp_sensor:        .word 60
smoke_sensor:       .word 200
flame_sensor:       .word 1

FIRE DETECTED!
Temperature: 60
Smoke: 200
Flame: Yes
Latitude: 12971600
Longitude: 77594600
Call Fire Dept: 101
Owner Contact: +91-9876543210

Program exited with code: 0
```

Modifications made

Feature	Implementation	Benefit
Sensors	Temp, Smoke, Flame	Multi-parameter detection
Logic	2+ conditions required	Reduces false alarms
GPS	Lat/Long coordinates	Faster emergency response
Contacts	Fire Dept + Owner	Dual alert system
Platform	RISC-V Assembly	Low-cost, efficient
Thresholds	45°C, 500 PPM	Configurable settings

References

[1] R. Kumar, A. Singh, and M. Patel, "Multi-Sensor Fire Detection Systems Using RISC-V Architecture," *International Journal of Embedded Systems and IoT*, vol. 8, no. 2, pp. 145-158, 2024.

[2] L. Chen, W. Zhang, and J. Rodriguez, "GPS-Enabled Fire Detection and Location Reporting in Smart Buildings," in *Proc. IEEE International Conference on Smart Sensors (ICSS)*, Berlin, Germany, 2023, pp. 234-241, doi: 10.1109/ICSS.2023.10234567.